Yoav Gudes

AI/ML & Semiconductor Engineering Co-op Student
University of Waterloo 2028
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Summary

- Waterloo Nanotechnology Engineering student (Nanoelectronics & Quantum Engineering) with 16 months of co-op.
- Built NLP classifiers and multi-step LLM pipelines in Python, from raw data and labeling to evaluation and production.
- Hardware depth from validating 50+ NVIDIA/AMD GPUs with CUDA and cleanroom fabrication of semiconductor devices.

Education

Nanotechnology Engineering: Nanoelectronics & Quantum Engineering Options | University of Waterloo **Sept 2023 – Apr 2028**
Relevant Coursework: Programming, Semiconductor Physics & Devices, Linear Algebra, Probability & Statistics

Work Experience

Process Engineering Intern | WOLF Advanced Technology **Sept 2025 – Apr 2026**

- Assembled and validated 50+ NVIDIA/AMD GPUs, benchmarking bandwidth (CUDA) and thermal stability (FurMark).
- Wired GPU modules for −45 °C to +85 °C thermal cycling, writing Python scripts to log die temperature data.
- Led stack-up failure analysis across 10 failed GPU units, tracing root causes to assembly and design issues.
- Debugged returned units through the RMA process, isolating and reworking faulty components, then re-validating.

AI/ML Engineering & Data Science Intern | TD Bank **Jan 2025 – Apr 2025**

- Developed an NLP sentiment classifier for client calls, building an Azure Databricks speech-to-text and parsing pipeline.
- Hand-labeled 10,000+ utterances; trained XGBoost on MPNet embeddings with class rebalancing and Optuna tuning.
- Benchmarked three model variants on precision, recall and F1, reaching up to 98.8% accuracy as an Azure AI alternative.

Data Engineering Intern | MedReddie **May 2024 – Aug 2024**

- Built a multi-step LLM pipeline (Gemini API) that found supplier websites, validated pages and extracted contacts.
- Collected data for 8,000+ suppliers and $50M+ in products, loading it into staging and production with SQL and Python.

Technical Skills – Hardware & Software

AI & Machine Learning

- *Models:* XGBoost, Random Forests, MPNet
- *LLMs:* Multi-Step Pipelines, Few-Shot Prompting
- *Agentic Tools:* Claude, Codex, MCP Connectors

Hardware Validation & Testing

- *GPU Validation:* CUDA & FurMark Stress Testing
- *Testing:* Thermal Cycling & Python Data Logging
- *Failure Analysis:* Root-Cause Debug & RMA Repair

Software & Data Engineering

- *Languages:* Python, SQL, Java, MATLAB
- *Libraries:* Pandas, scikit-learn, Optuna, Selenium
- *Platforms:* Azure Databricks, Jupyter, Git

Semiconductor Devices & Fabrication

- *Devices:* Diodes, MIS Capacitors, MOSFETs & CMOS
- *Fabrication:* PECVD, Lithography, RIE & Sputtering
- *Characterization:* C–V/I–V, Probe Station & SEM

Leadership & Achievements

Student Body President | Hillel Waterloo & Laurier **May 2025 – Aug 2026**

- Led student programming and oversaw executive board operations across a multi-campus organization.

Awards

- **University of Waterloo:** Term Distinction (Dean's List), Excellent Academic Standing & Co-op Review, Jordan Banks Award.
- **TD Bank:** Achieve the Dream Canada Scholarship, Co-op Challenge Finalist
- **Hillel Ontario:** Lewis and Olga Boros Scholarship

Languages: English & Hebrew (Fluent); Portuguese & Spanish (Conversational)